Supplementary Material: Proofs of the Four Propositions

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We use $\tilde{Q}_i = x_i^E \alpha_i Q_i + x_i^F Q_i$ and $\tilde{T} = \sum_i \tilde{Q}_i$ for the effective contribution aggregate in Eq. (15). Let $d_i = (1-\alpha_i)Q_i$ and

$$S_{-i} = \sum_{j \neq i} x_j^F d_j.$$

The private utilities are

$$U_i(N) = -v_i L(S_{-i}),$$

$$U_i(E) = P_i(E) - c_{i,1}\alpha_i Q_i - v_i L(S_{-i}),$$

$$U_i(F) = P_i(F) - c_{i,1}\alpha_i Q_i - v_i L(S_{-i} + d_i).$$

Define

$$\Delta P_i = P_i(F) - P_i(E), \qquad \Delta C_i(S_{-i}) = v_i[L(S_{-i} + d_i) - L(S_{-i})].$$

Proposition 1

Participation

$$U_i(E) - U_i(N) = P_i(E) - c_{i,1}\alpha_i Q_i.$$

Hence

$$U_i(E) \geq U_i(N) \iff P_i(E) \geq c_{i,1}\alpha_i Q_i.$$

Upgrade

$$U_i(F) - U_i(E) = P_i(F) - P_i(E) - v_i[L(S_{-i} + d_i) - L(S_{-i})] = \Delta P_i - \Delta C_i(S_{-i}).$$

Thus

$$U_i(F) \geq U_i(E) \iff \Delta P_i \geq \Delta C_i(S_{-i}).$$

If $d_i > 0$, Proposition 3 gives that $\Delta C_i(S_{-i})$ is decreasing in S_{-i}. Therefore, if

$$\Delta P_i = \Delta C_i(\bar{S}_i),$$

then

$$S_{-i} < \bar{S}_i \Rightarrow U_i(E) > U_i(F),$$

and

$$S_{-i} \geq \bar{S}_i \Rightarrow U_i(F) \geq U_i(E).$$

This is the conditional E/F threshold, with the final action still selected from $\{N, E, F\}$. $\square$

Proposition 2

Let

$$W^{soc}(\mathbf{a}) = P(\tilde{T}(\mathbf{a})) - \sum_{j \in \mathcal{N}} C_j^{soc}(a_j, S_{-j}, \mathbf{a}_{-j}),$$

where

$$C_i^{soc} = C_i^{priv} + \eta E_i^{caused}.$$

For an $E \to F$ deviation by member i,

$$\Delta W_i^{E \to F} = W^{soc}(F_i, \mathbf{a}_{-i}) - W^{soc}(E_i, \mathbf{a}_{-i}).$$

By direct substitution,

$$\Delta W_i^{E \to F} = \underbrace{P(\tilde{T}(F_i, \mathbf{a}_{-i})) - P(\tilde{T}(E_i, \mathbf{a}_{-i}))}_{\Delta R_i^{soc}} - \underbrace{\sum_{j \in \mathcal{N}} \left[C_j^{soc}(F_i, \mathbf{a}_{-i}) - C_j^{soc}(E_i, \mathbf{a}_{-i})\right]}_{\Delta C_i^{soc}}.$$

Hence

$$\Delta W_i^{E \to F} \geq 0 \iff \Delta R_i^{soc} \geq \Delta C_i^{soc}.$$

If

$$\eta \Delta E_i^{caused}(\mathbf{a}_{-i}) > 0,$$

then

$$\Delta C_i^{soc} = \Delta C_i^{priv}(S_{-i}) + \eta \Delta E_i^{caused}(\mathbf{a}_{-i}) > \Delta C_i^{priv}(S_{-i}).$$

Therefore, the interval

$$\Delta C_i^{priv}(S_{-i}) \leq \Delta P_i < \Delta C_i^{soc}$$

is a socially excessive full-sharing region: F is privately optimal but not socially welfare-improving. $\square$

Proposition 3

Assume $L'(S) > 0$, $L''(S) < 0$, and $d_i > 0$. Then

$$\Delta C_i(S_{-i}) = v_i[L(S_{-i} + d_i) - L(S_{-i})],$$

so

$$\frac{\partial \Delta C_i(S_{-i})}{\partial S_{-i}} = v_i[L'(S_{-i} + d_i) - L'(S_{-i})].$$

Since $L''(S) < 0$, $L'(S)$ is strictly decreasing. Because $d_i > 0$,

$$L'(S_{-i} + d_i) < L'(S_{-i}),$$

and therefore

$$\frac{\partial \Delta C_i(S_{-i})}{\partial S_{-i}} < 0.$$

If $d_i = 0$, then $\Delta C_i(S_{-i}) \equiv 0$. For $L(S) = a(1 - e^{-bS})$,

$$\Delta C_i(S_{-i}) = v_i a e^{-bS_{-i}}(1 - e^{-bd_i}),$$

which is decreasing in S_{-i} whenever $d_i > 0$. $\square$

Proposition 4

Let

$$\mathcal{A}(\mathbf{a}) = \{i : a_i \in \{E, F\}\}$$

be the participating set. Non-participants receive $U_i(N, \mathbf{a}_{-i})$ and no cooperative transfer. For $i \in \mathcal{A}$, use the corrected redistribution transfer

$$\tilde{x}_i^R = (1 - \beta)\frac{\tilde{Q}_i}{\tilde{T}}P(\tilde{T}) + \beta C_i^{priv}(a_i, S_{-i}) + k\mathbf{1}\{i \in \mathcal{K}\}.$$

Here

$$C_{total}^{priv} = \sum_{i \in \mathcal{A}} C_i^{priv}(a_i, S_{-i}).$$

Budget balance

Summing over $i \in \mathcal{A}$,

$$\sum_{i \in \mathcal{A}} \tilde{x}_i^R = (1-\beta)P(\tilde{T}) \sum_{i \in \mathcal{A}} \frac{\tilde{Q}_i}{\tilde{T}} + \beta C_{total}^{priv} + k|\mathcal{K}|.$$

Since

$$\sum_{i \in \mathcal{A}} \frac{\tilde{Q}_i}{\tilde{T}} = 1,$$

budget balance,

$$\sum_{i \in \mathcal{A}} \tilde{x}_i^R = P(\tilde{T}),$$

requires

$$(1-\beta)P(\tilde{T}) + \beta C_{total}^{priv} + k|\mathcal{K}| = P(\tilde{T}).$$

Thus

$$k = \frac{\beta[P(\tilde{T}) - C_{total}^{priv}]}{|\mathcal{K}|}.$$

If $k \geq 0$ is required, then $P(\tilde{T}) \geq C_{total}^{priv}$.

Critical β

For $i \in \mathcal{K}$, define

$$B_i = \frac{\tilde{Q}_i}{\tilde{T}} P(\tilde{T}) - C_i^{priv}(a_i, S_{-i}),$$

and

$$R_K = \frac{P(\tilde{T}) - C_{total}^{priv}}{|\mathcal{K}|}.$$

Using the expression for k,

$$U_i^R = (1-\beta)B_i + \beta R_K.$$

The binding individual-rationality constraint

$$U_i^R = \pi_i^{solo}$$

gives

$$(1-\beta)B_i + \beta R_K = \pi_i^{solo},$$

so, if $R_K \neq B_i$,

$$\beta_{0,i} = \frac{\pi_i^{solo} - B_i}{R_K - B_i}.$$

Local high-externality state

Consider a candidate state $\mathbf{a}^H$ with $a_i^H = F$ for all $i \in \mathcal{M}$. If

$$\Delta P_i \geq \Delta C_i(S_{-i}^H)$$

and

$$U_i(F; \mathbf{a}_{-i}^H) \geq U_i(N; \mathbf{a}_{-i}^H)$$

for every $i \in \mathcal{M}$, then no such member benefits by deviating from F to E or N. The state is locally stable on $\mathcal{M}$.

Feasibility shrinkage

If externalities raise C_{total}^{priv}, then

$$R_K = \frac{P(\tilde{T}) - C_{total}^{priv}}{|\mathcal{K}|}$$

falls. If at least one core member requires $\beta_{0,i} \notin [0,1]$, no admissible one-dimensional β can simultaneously satisfy budget balance and that member's individual-rationality constraint. This is the failure region of the parameterized redistribution mechanism. □